ID3 worked example

Exercise

Build a decision tree with ID3 for the following training set of e-mails, where $x_1 = \texttt{free}$, $x_2 = \texttt{winner}$, $x_3 = \texttt{meeting}$ indicate whether the term appears in the e-mail, and $y \in \{\texttt{S}, \texttt{L}\}$ is the class ($\texttt{S}$ = spam, $\texttt{L}$ = legitimate).

ID	x_1	x_2	x_3	y
m_1	T	F	F	S
m_2	F	F	T	L
m_3	T	T	T	S
m_4	F	F	F	L
m_5	F	T	F	S
m_6	T	F	T	L
m_7	F	F	F	L
m_8	T	T	F	S
m_9	F	F	F	L
m_{10}	T	F	F	S

Solution

At each node ID3 evaluates every remaining attribute X on the examples reaching that node and chooses the one with the highest

$$\text{InformationGain}(X) = H(Y) - H(Y \mid X),$$

where

$$H(Y) = -\sum_i P(Y = y_i) \log_2 P(Y = y_i)$$

is the entropy of the class label (how uncertain the class is, in bits) and

$$H(Y \mid X) = \sum_v P(X = v)\, H(Y \mid X = v)$$

is the entropy that remains once the value of X is known (with the convention $0 \log_2 0 = 0$).

For each binary attribute we first compute the probabilities $P(X = v)$ and $P(Y = y \mid X = v)$ from the table of the node, and then substitute them in the entropy equations.

Step 0: sort the examples by label

ID	x_1	x_2	x_3	y
m_1	T	F	F	S
m_3	T	T	T	S
m_5	F	T	F	S
m_8	T	T	F	S
m_{10}	T	F	F	S
m_2	F	F	T	L
m_4	F	F	F	L
m_6	T	F	T	L
m_7	F	F	F	L
m_9	F	F	F	L

Node 1 (root): all 10 examples

Attribute	Value	Samples	Labels
x_1	T	m_1, m_3, m_6, m_8, m_{10}	S, S, L, S, S
	F	m_2, m_4, m_5, m_7, m_9	L, L, S, L, L
x_2	T	m_3, m_5, m_8	S, S, S
	F	m_1, m_2, m_4, m_6, m_7, m_9, m_{10}	S, L, L, L, L, L, S
x_3	T	m_2, m_3, m_6	L, S, L
	F	m_1, m_4, m_5, m_7, m_8, m_9, m_{10}	S, L, S, L, S, L, S

Entropy of the class. 5 S, 5 L:

$$P(Y = \texttt{S}) = P(Y = \texttt{L}) = \tfrac{5}{10} = 0.5, \qquad H(Y) = -0.5 \log_2 0.5 - 0.5 \log_2 0.5 = 1.$$

Attribute x_1 (T: $m_1, m_3, m_6, m_8, m_{10}$; F: m_2, m_4, m_5, m_7, m_9)

$$\begin{aligned}
P(X_1 = \texttt{T}) &= \tfrac{5}{10} = 0.5 \\
P(X_1 = \texttt{F}) &= \tfrac{5}{10} = 0.5 \\
P(Y = \texttt{S} \mid X_1 = \texttt{T}) &= \tfrac{4}{5} = 0.8 \\
P(Y = \texttt{L} \mid X_1 = \texttt{T}) &= \tfrac{1}{5} = 0.2 \\
P(Y = \texttt{S} \mid X_1 = \texttt{F}) &= \tfrac{1}{5} = 0.2 \\
P(Y = \texttt{L} \mid X_1 = \texttt{F}) &= \tfrac{4}{5} = 0.8
\end{aligned}$$

$$\begin{aligned}
H(Y \mid X_1 = \texttt{T}) &= -0.8 \log_2 0.8 - 0.2 \log_2 0.2 \approx 0.72 \\
H(Y \mid X_1 = \texttt{F}) &= -0.2 \log_2 0.2 - 0.8 \log_2 0.8 \approx 0.72 \\
H(Y \mid X_1) &= 0.5 \cdot 0.72 + 0.5 \cdot 0.72 = 0.72 \\
\text{InformationGain}(X_1) &= 1 - 0.72 = \mathbf{0.28}
\end{aligned}$$

Attribute x_2 (T: m_3, m_5, m_8; F: $m_1, m_2, m_4, m_6, m_7, m_9, m_{10}$)

$$\begin{aligned}
P(X_2 = \texttt{T}) &= \tfrac{3}{10} = 0.3 \\
P(X_2 = \texttt{F}) &= \tfrac{7}{10} = 0.7 \\
P(Y = \texttt{S} \mid X_2 = \texttt{T}) &= \tfrac{3}{3} = 1 \\
P(Y = \texttt{L} \mid X_2 = \texttt{T}) &= \tfrac{0}{3} = 0 \\
P(Y = \texttt{S} \mid X_2 = \texttt{F}) &= \tfrac{2}{7} \\
P(Y = \texttt{L} \mid X_2 = \texttt{F}) &= \tfrac{5}{7}
\end{aligned}$$

$$H(Y \mid X_2 = \texttt{T}) = -1 \log_2 1 - 0 \log_2 0 = 0$$
$$H(Y \mid X_2 = \texttt{F}) = -\tfrac{2}{7} \log_2 \tfrac{2}{7} - \tfrac{5}{7} \log_2 \tfrac{5}{7} \approx 0.86$$
$$H(Y \mid X_2) = 0.3 \cdot 0 + 0.7 \cdot 0.86 = 0.60$$
$$\text{InformationGain}(X_2) = 1 - 0.60 = \mathbf{0.40}$$

Attribute x_3 (T: m_2, m_3, m_6; F: $m_1, m_4, m_5, m_7, m_8, m_9, m_{10}$)

$$P(X_3 = \texttt{T}) = \tfrac{3}{10} = 0.3$$
$$P(X_3 = \texttt{F}) = \tfrac{7}{10} = 0.7$$
$$P(Y = \texttt{S} \mid X_3 = \texttt{T}) = \tfrac{1}{3}$$
$$P(Y = \texttt{L} \mid X_3 = \texttt{T}) = \tfrac{2}{3}$$
$$P(Y = \texttt{S} \mid X_3 = \texttt{F}) = \tfrac{4}{7}$$
$$P(Y = \texttt{L} \mid X_3 = \texttt{F}) = \tfrac{3}{7}$$

$$H(Y \mid X_3 = \texttt{T}) = -\tfrac{1}{3} \log_2 \tfrac{1}{3} - \tfrac{2}{3} \log_2 \tfrac{2}{3} \approx 0.92$$
$$H(Y \mid X_3 = \texttt{F}) = -\tfrac{4}{7} \log_2 \tfrac{4}{7} - \tfrac{3}{7} \log_2 \tfrac{3}{7} \approx 0.99$$
$$H(Y \mid X_3) = 0.3 \cdot 0.92 + 0.7 \cdot 0.99 = 0.97$$
$$\text{InformationGain}(X_3) = 1 - 0.97 = \mathbf{0.03}$$

Choice. x_2 has the highest gain, so it is the root. $x_2 = \texttt{T}$ contains only spam: leaf S. $x_2 = \texttt{F}$ is mixed and is expanded (Node 2).

Branch	ID	x_1	x_2	x_3	y
$x_2 = \texttt{T}$	m_3	T	T	T	S
	m_5	F	T	F	S
	m_8	T	T	F	S
$x_2 = \texttt{F}$	m_1	T	F	F	S
	m_2	F	F	T	L
	m_4	F	F	F	L
	m_6	T	F	T	L
	m_7	F	F	F	L
	m_9	F	F	F	L
	m_{10}	T	F	F	S

Node 2: branch $x_2 = \texttt{F}$ (7 examples)

Attribute	Value	Samples	Labels
x_1	T	m_1, m_6, m_{10}	S, L, S
	F	m_2, m_4, m_7, m_9	L, L, L, L
x_3	T	m_2, m_6	L, L
	F	m_1, m_4, m_7, m_9, m_{10}	S, L, L, L, S

Entropy of the class. 2 S, 5 L:

$$P(Y = \texttt{S} \mid X_2 = \texttt{F}) = \tfrac{2}{7},\ P(Y = \texttt{L} \mid X_2 = \texttt{F}) = \tfrac{5}{7}, \qquad H(Y \mid X_2 = \texttt{F}) = -\tfrac{2}{7} \log_2 \tfrac{2}{7} - \tfrac{5}{7} \log_2 \tfrac{5}{7} \approx 0.86.$$

Attribute x_1 (T: m_1, m_6, m_{10}; F: m_2, m_4, m_7, m_9)

$$P(X_1 = \texttt{T} \mid X_2 = \texttt{F}) = \tfrac{3}{7}$$
$$P(X_1 = \texttt{F} \mid X_2 = \texttt{F}) = \tfrac{4}{7}$$
$$P(Y = \texttt{S} \mid X_1 = \texttt{T}, X_2 = \texttt{F}) = \tfrac{2}{3}$$
$$P(Y = \texttt{L} \mid X_1 = \texttt{T}, X_2 = \texttt{F}) = \tfrac{1}{3}$$
$$P(Y = \texttt{S} \mid X_1 = \texttt{F}, X_2 = \texttt{F}) = \tfrac{0}{4} = 0$$
$$P(Y = \texttt{L} \mid X_1 = \texttt{F}, X_2 = \texttt{F}) = \tfrac{4}{4} = 1$$

$$H(Y \mid X_1 = \texttt{T}, X_2 = \texttt{F}) = -\tfrac{2}{3}\log_2\tfrac{2}{3} - \tfrac{1}{3}\log_2\tfrac{1}{3} \approx 0.92$$
$$H(Y \mid X_1 = \texttt{F}, X_2 = \texttt{F}) = -0\log_2 0 - 1\log_2 1 = 0$$
$$H(Y \mid X_1, X_2 = \texttt{F}) = \tfrac{3}{7}\cdot 0.92 + \tfrac{4}{7}\cdot 0 \approx 0.39$$
$$\text{InformationGain}(X_1) = 0.86 - 0.39 = \mathbf{0.47}$$

Attribute x_3 (T: m_2, m_6; F: $m_1, m_4, m_7, m_9, m_{10}$)

$$P(X_3 = \texttt{T} \mid X_2 = \texttt{F}) = \tfrac{2}{7}$$
$$P(X_3 = \texttt{F} \mid X_2 = \texttt{F}) = \tfrac{5}{7}$$
$$P(Y = \texttt{S} \mid X_3 = \texttt{T}, X_2 = \texttt{F}) = \tfrac{0}{2} = 0$$
$$P(Y = \texttt{L} \mid X_3 = \texttt{T}, X_2 = \texttt{F}) = \tfrac{2}{2} = 1$$
$$P(Y = \texttt{S} \mid X_3 = \texttt{F}, X_2 = \texttt{F}) = \tfrac{2}{5} = 0.4$$
$$P(Y = \texttt{L} \mid X_3 = \texttt{F}, X_2 = \texttt{F}) = \tfrac{3}{5} = 0.6$$

$$H(Y \mid X_3 = \texttt{T}, X_2 = \texttt{F}) = -0\log_2 0 - 1\log_2 1 = 0$$
$$H(Y \mid X_3 = \texttt{F}, X_2 = \texttt{F}) = -0.4\log_2 0.4 - 0.6\log_2 0.6 \approx 0.97$$
$$H(Y \mid X_3, X_2 = \texttt{F}) = \tfrac{2}{7}\cdot 0 + \tfrac{5}{7}\cdot 0.97 \approx 0.69$$
$$\text{InformationGain}(X_3) = 0.86 - 0.69 = \mathbf{0.17}$$

Choice. x_1 has the highest gain. $x_1 = \texttt{F}$ contains only legitimate e-mails: leaf L. $x_1 = \texttt{T}$ is mixed and is expanded (Node 3).

Branch	ID	x_1	x_2	x_3	y
$x_1 = \texttt{T}$	m_1	T	F	F	S
	m_6	T	F	T	L
	m_{10}	T	F	F	S
$x_1 = \texttt{F}$	m_2	F	F	T	L
	m_4	F	F	F	L
	m_7	F	F	F	L
	m_9	F	F	F	L

Node 3: branch $x_2 = \texttt{F},\ x_1 = \texttt{T}$ (3 examples)

Attribute	Value	Samples	Labels
x_3	T	m_6	L
	F	m_1, m_{10}	S, S

Entropy of the class. 2 S, 1 L: $H(Y \mid X_1 = \texttt{T}, X_2 = \texttt{F}) = -\tfrac{2}{3}\log_2\tfrac{2}{3} - \tfrac{1}{3}\log_2\tfrac{1}{3} \approx 0.92$.

Attribute x_3 (the only one left; `T`: m_6; `F`: m_1, m_{10})

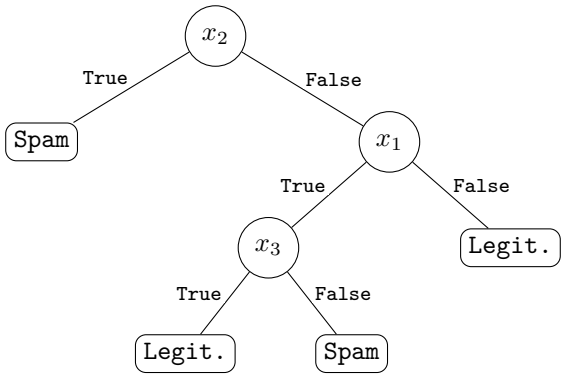
$$
\begin{aligned}
P(X_3 = \texttt{T} \mid \cdot) &= \tfrac{1}{3} \\
P(X_3 = \texttt{F} \mid \cdot) &= \tfrac{2}{3} \\
P(Y = \texttt{S} \mid X_3 = \texttt{T}, \cdot) &= \tfrac{0}{1} = 0 \\
P(Y = \texttt{L} \mid X_3 = \texttt{T}, \cdot) &= \tfrac{1}{1} = 1 \\
P(Y = \texttt{S} \mid X_3 = \texttt{F}, \cdot) &= \tfrac{2}{2} = 1 \\
P(Y = \texttt{L} \mid X_3 = \texttt{F}, \cdot) &= \tfrac{0}{2} = 0
\end{aligned}
$$

$$
\begin{aligned}
H(Y \mid X_3 = \texttt{T}, \cdot) &= -0 \log_2 0 - 1 \log_2 1 = 0 \\
H(Y \mid X_3 = \texttt{F}, \cdot) &= -1 \log_2 1 - 0 \log_2 0 = 0 \\
H(Y \mid X_3, \cdot) &= \tfrac{1}{3} \cdot 0 + \tfrac{2}{3} \cdot 0 = 0 \\
\text{InformationGain}(X_3) &= 0.92 - 0 = \mathbf{0.92}
\end{aligned}
$$

where $\cdot$ stands for $X_1 = \texttt{T}, X_2 = \texttt{F}$. Both children are pure, so they are leaves.

Branch	ID	x_1	x_2	x_3	y
$x_3 = \texttt{T}$	m_6	T	F	T	L
$x_3 = \texttt{F}$	m_1	T	F	F	S
	m_{10}	T	F	F	S

Resulting tree



Equivalent rules: IF `winner` THEN spam; IF ¬`winner` AND `free` AND ¬`meeting` THEN spam; otherwise legitimate. The tree classifies all 10 training e-mails correctly.